
# Advanced Cisco Cybersecurity Academy – FortiGate-40F Factory Reset and Initial SOHO Configuration

Yuuki Burleigh CCNA

**Purpose**

This lab serves to develop a basic understanding and familiarity with Fortinet technologies and systems, and more specifically the FortiGate-40F. This lab provides the basic skillset for navigating the FortiGate-40F GUI and making basic SOHO (Small Office Home Office) level configurations. Understanding how to use Fortinet hardware, software, and systems serve as a valuable skillset as they hold 55% of unit market share in firewalls as of late 2025. This lab helps to develop those skills and a deeper understanding of concepts like network access and network security.

**Background**

Fortinet is a global leader in cybersecurity solutions, founded in 200 and headquartered in Sunnyvale, California. They occupy approximately 3.93% of the total market share within the entire IT field, and over half the market in firewall shipments. Because of these facts, HR and IT-decision makers value Fortinet certifications and skills highly, with 89% of leaders preferring to hire certified candidates to prove their skills with Fortinet Firewalls and systems. Fortinet has been able to be one of the top choices for IT solutions due to its AI-driven Security Fabric Architecture that integrates over 50 enterprise grade products to provide easier and enhanced central management systems, where less people can more easily and effectively manage larger and more complex architectures that incorporate multiple separate Fortinet products. Fortinet additionally is the only cybersecurity solutions company to use SPU's or Security Processing Units to give them an edge over their competition; these units work more efficiently and can handle higher workload when it comes to tasks related to cybersecurity compared to normal CPUs resulting in higher efficiency, performance, and low latency. These special advantages that Fortinet systems utilize have driven skills related to Fortinet to become increasingly valuable within the IT field.

The FortiGate-40F and all Fortinet devices come with the FortiOS. FortiOS is Fortinet's natively AI-powered and Quantum Safe operating system and serves as the foundation for converging networking and security to create a unified security platform. This approach allows Fortinet systems to eliminate any possible gaps in security created by individual devices and products by converging everything into one unified architecture.  This way of managing security through a centralized security platform is a Security Fabric. Fortinet uses their Operating System to converge everything into a security fabric to then leverage AI to monitor the entire network, allowing for devices like the FortiGate-40F to provide high-level security to a network. This lab uses the FortiGate-40F to provide secure internet access to wired end devices in a SOHO environment. The FortiGate-40F can also be used to create a wireless network to access the internet without a physical connection but will be covered in a later lab. A SOHO environment is an environment where one to ten users are accommodated usually through one router/firewall, in this case a FortiGate-40F. This includes almost every household in the US and the millions of small businesses who all require a secure way to access the internet. Most of these households and businesses use ISP-supplied routers and gateways, and Parks Associates found that 71% of U.S. internet households lease, purchase, or receive routers from their ISP.

This sometimes is not a viable solution for businesses who require privacy and households who may include remote workers that deal with sensitive information, as ISP bought and rented routers can access all information that goes in and out of the network. This is where being able to configure a SOHO network using a firewall like the FortiGate-40F becomes a valuable skill to have. Additionally, being able to further isolate a SOHO network from an ISP's control allows for a significantly reduced chance of any sensitive information being leaked during attacks done on major ISPs. An example of SOHO networks getting attacked due to an ISP vulnerability is an attack that compromised 600,000 SOHO routers within the span of three days using a trojan to install remote access software using a firmware update as a guise. This attack only affected ISP issued routers, but isolating your device from an ISP does not mean the device cannot be comprised. In 2021 61% of Small Businesses reported they received a cyberattack this is where the skills of being able to configure a strong SOHO network become valuable to prevent data breaches that can lead to lawsuits and sensitive information being leaked.

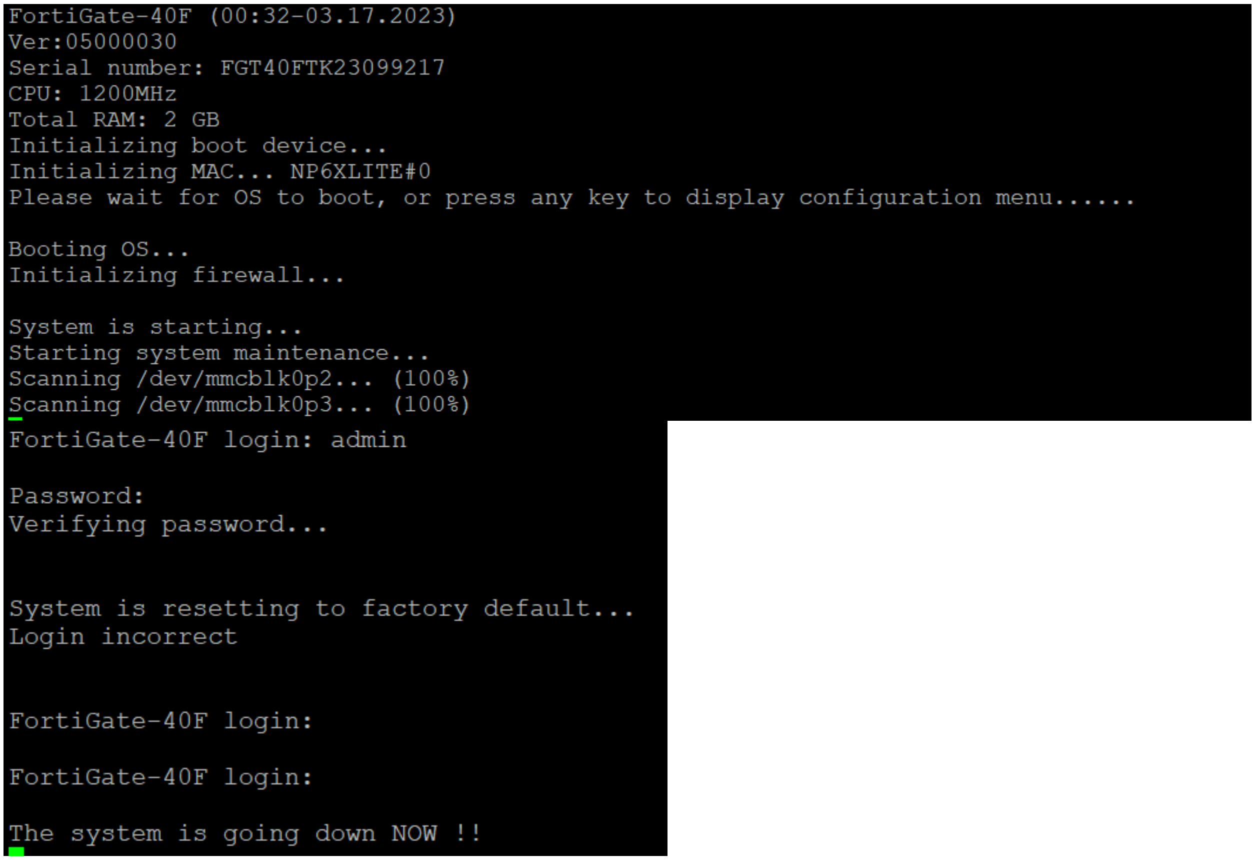
To configure a strong network using the FortiGate-40F a factory reset is needed as this will reset and previous configuration or unknown configuration that could contain spyware or malware. Once the firewall is reset, the Firewall should allow for internet access with its default factory configurations. To create a strong network, access control policies and internet policies must be implemented. The access control polices control who is allowed to access the network and internet, this prevents anyone without the proper credentials from accessing the firewall's network from the inside of the network. Internet policies allow for control over what can be accessed on the internet, on the FortiGate-40F separate polices can be made on what can be searched, downloaded, and seen on websites. This allows for granular control on what can be done using the internet preventing intentional or accidental download of malware on end devices within the network.

**Lab Summary**

In this lab the FortiGate-40F is factory reset through using the factory reset button physically on the firewall rather than using the CLI. Admin credentials were configured along with a local user. Updates were configured to take place as often as possible and done automatically. Then as the FortiGate-40F Firewall allows for internet access with default configuration, only policy adjustments to increase security on the default outbound policy were made.

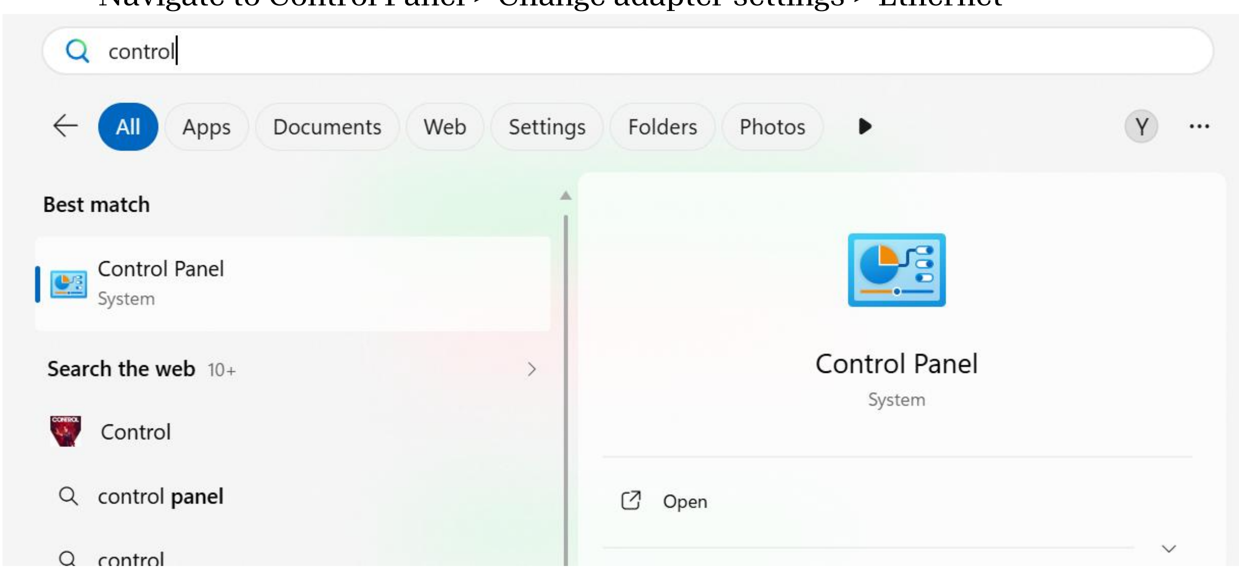
**Lab Commands**

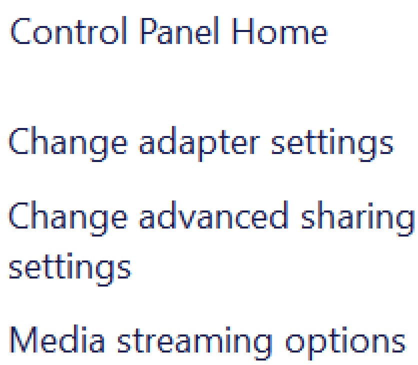
To Factory Reset the Firewall the firewall must be plugged in and the user will have to be plugged into the firewall via console cable and let the firewall boot completely then have the Factory Reset button pressed within 60 seconds of booting.

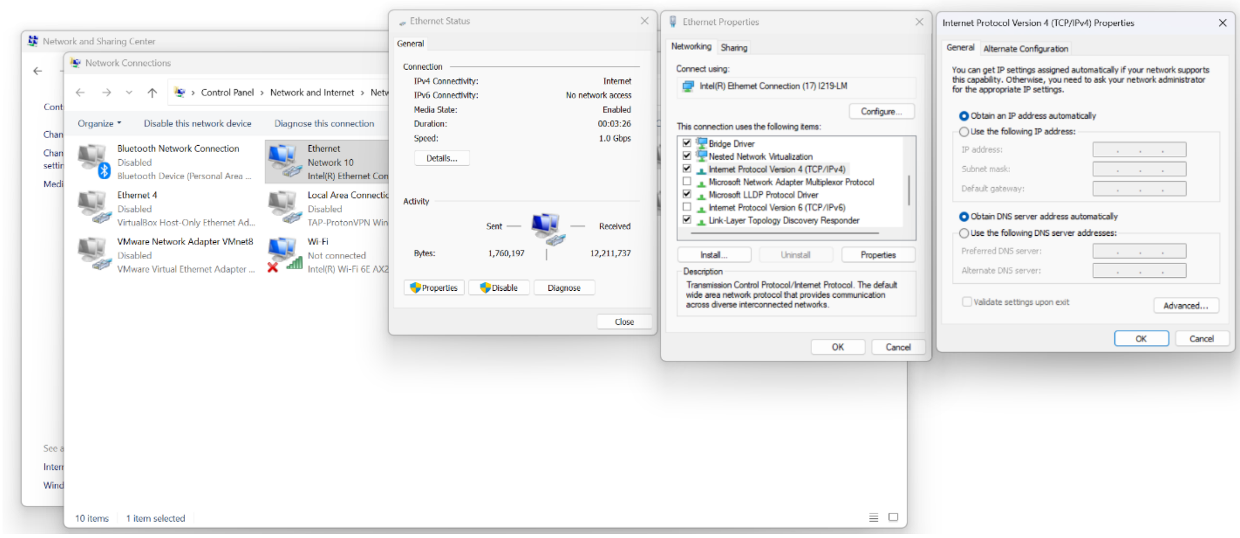



To then access the Firewall via GUI the device connecting to the firewall being be connected to the firewall's LAN 1 port and have IP address and default gateway be automatically configured by the firewall.

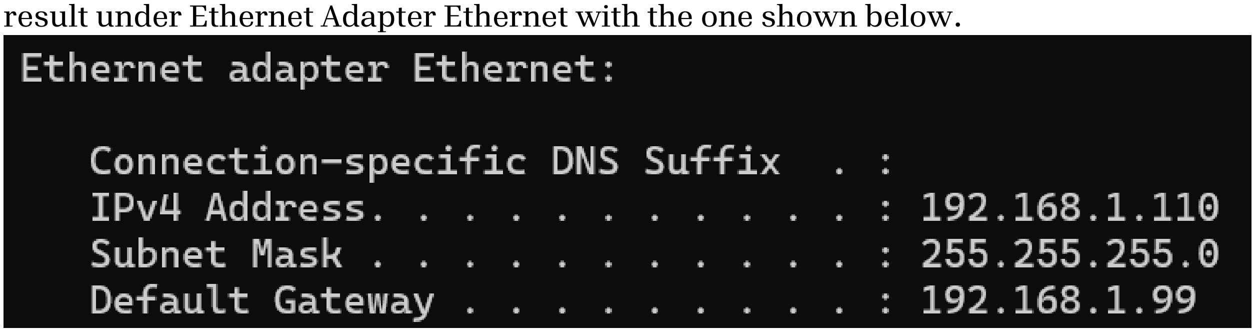
Navigate to Control Panel > Change adapter settings > Ethernet





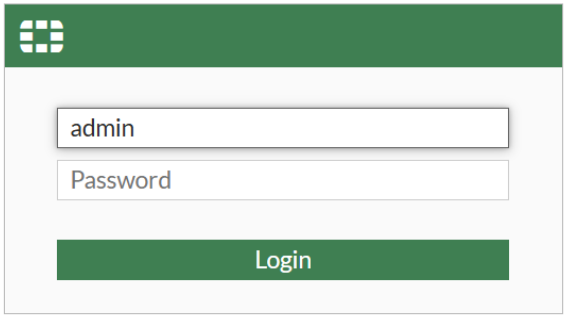



To ensure this is done correctly and the device connecting to the firewall has to proper IP addresses press the windows key and open command prompt by typing “cmd” and hitting enter. Then, type the command “ipconfig” and cross check your result under Ethernet Adapter Ethernet with the one shown below.

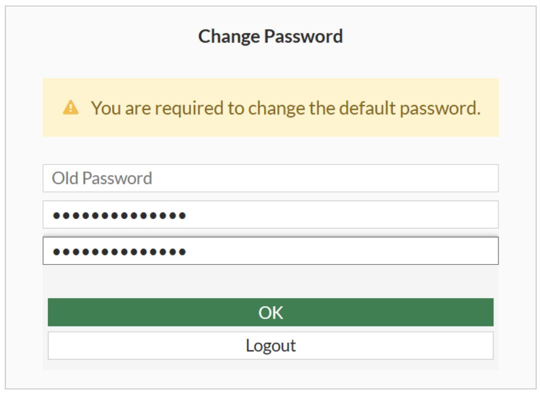


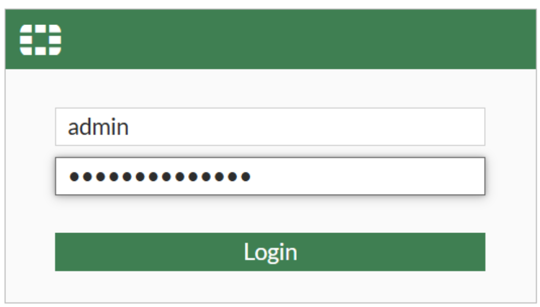

If this is done correctly then in a web browser type [https://192.168.1.99](https://192.168.1.99) to access the GUI.


Then you will be prompted to enter a username and password. Ignore the password field and only enter “admin” as your username and login.

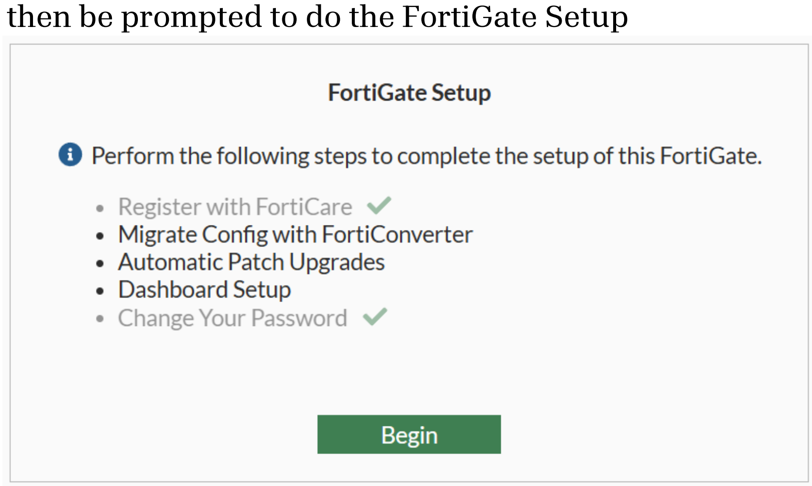



You will then be prompted to enter a password and will need to re-login with the new password.

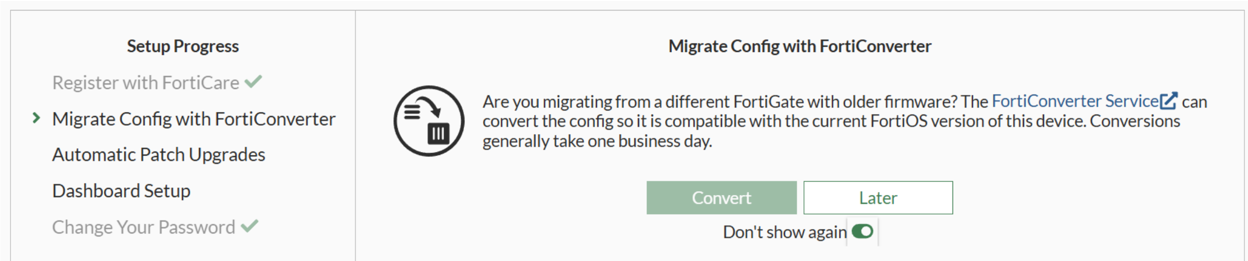


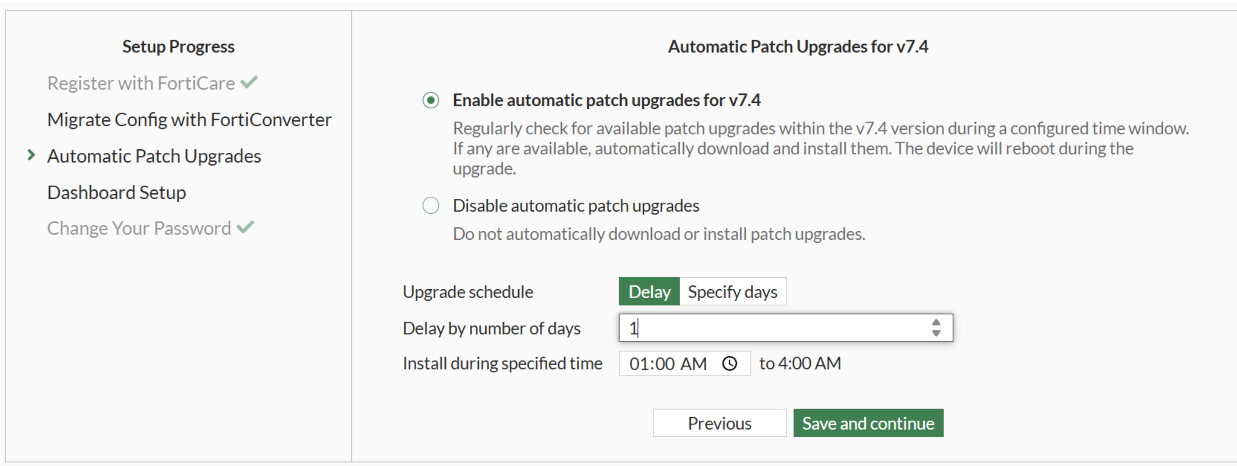

You will then be prompted to do the FortiGate Setup



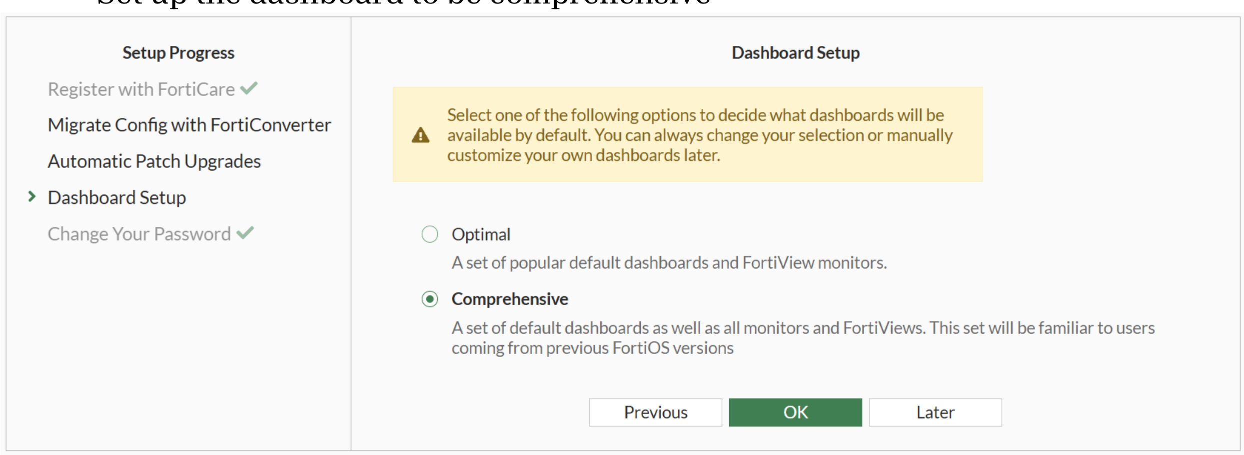
Do not migrate from an older firmware unless needed.



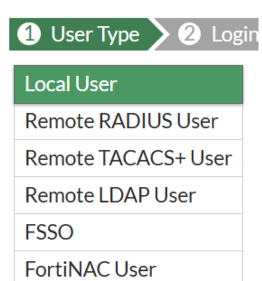

Configure the automatic patch upgrade to happen as often as possible



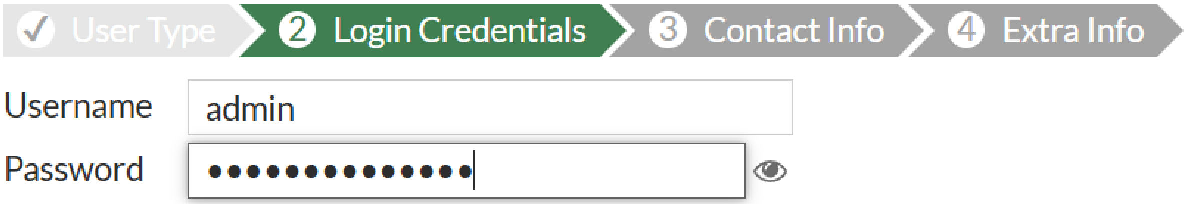

Set up the dashboard to be comprehensive



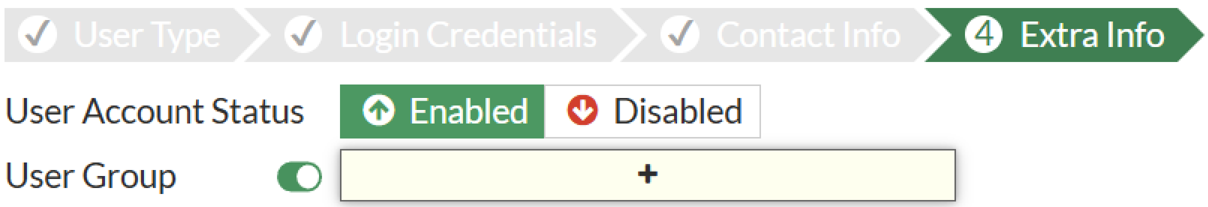

Now you should be shown the dashboard, now navigate to the left navigation panel and open User and Authentication and go to User Definition and create a local user.

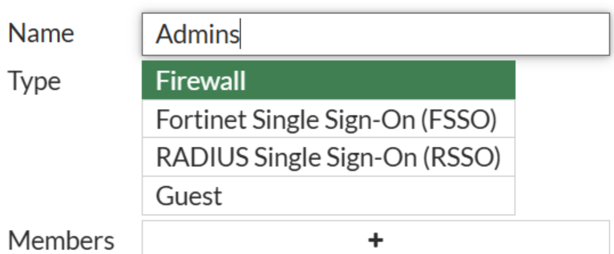


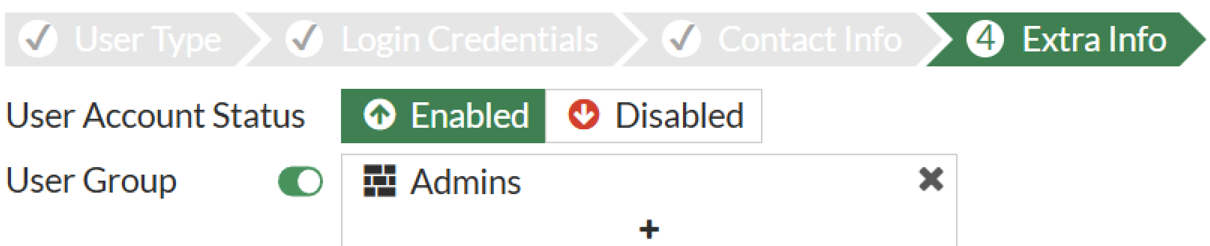
Create the login credentials




Skip the contact info unless you wanted, in Extra Info create a user group named Admins for the user.

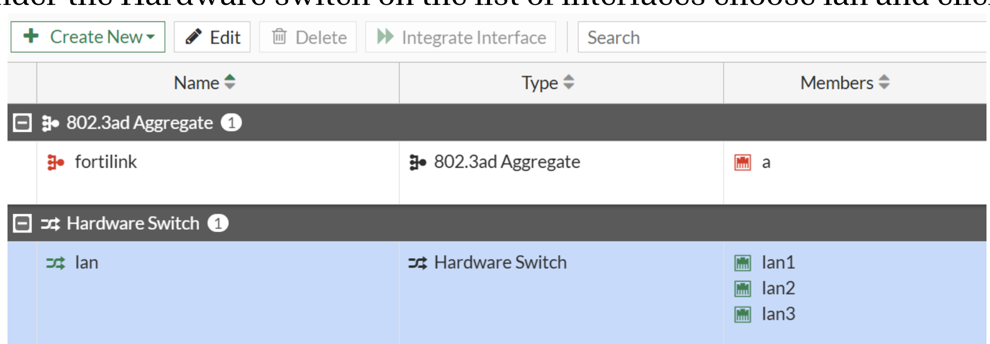


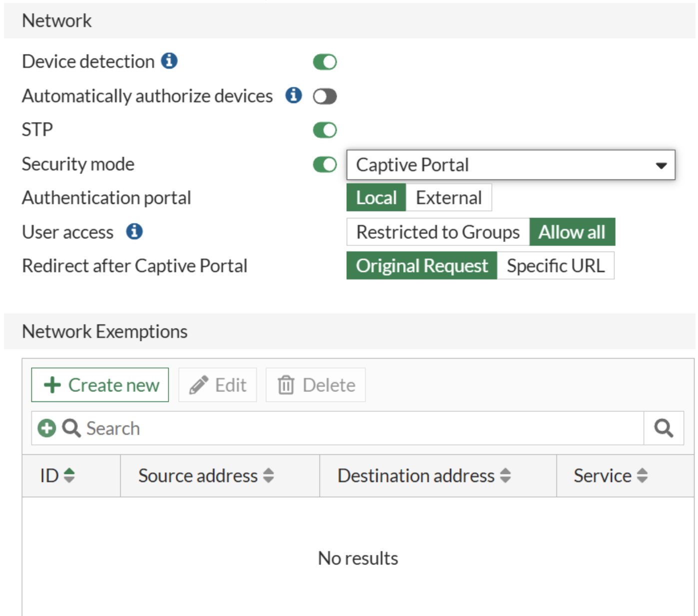




Finish creating the User and then navigate back to the navigation panel and go to Network > Interfaces.

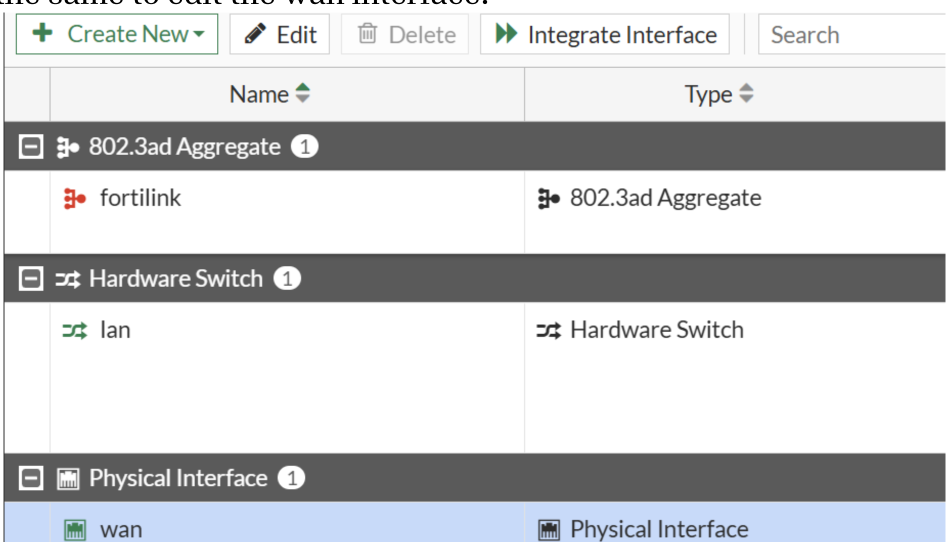
Under the Hardware switch on the list of interfaces choose lan and click edit.



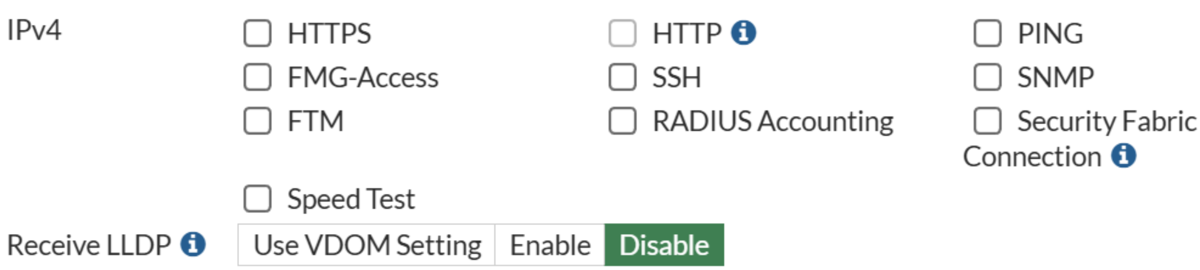
Then under the network settings for the LAN interfaces copy the setting below.




Do the same to edit the wan interface.

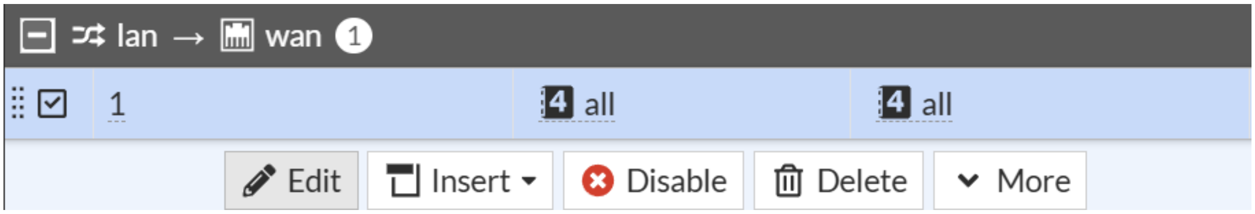



Edit to match the setting below, this will prevent anyone outside from connecting to the admin interface through the wan interface.

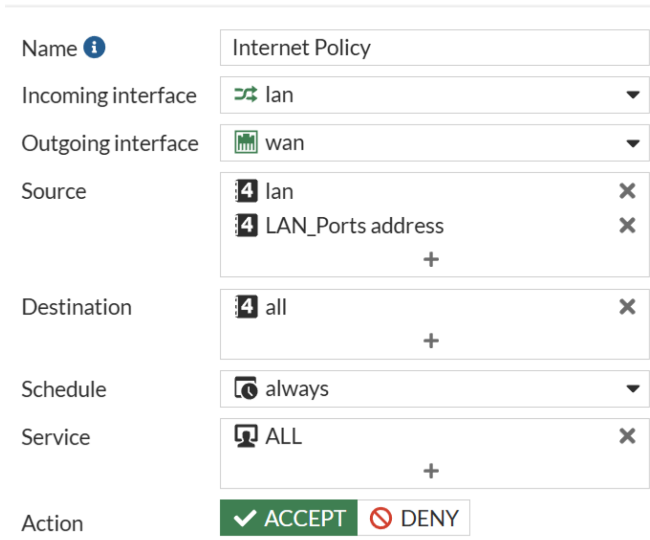



Confirm the configurations and navigate using the lefthand panel to Policy & Objects > Firewall Policy.

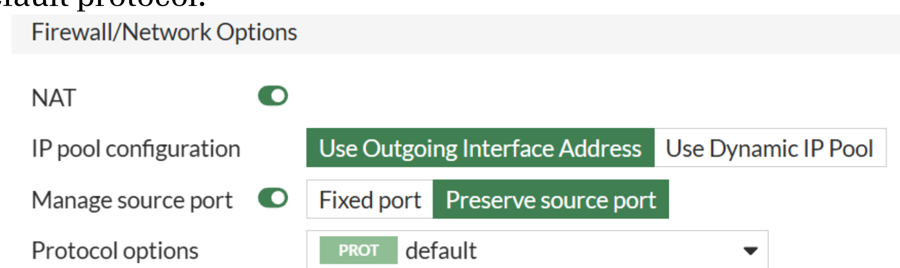
Select the policy named “1” that allows for traffic from the lan to go to the wan interface and be routed to the internet and select edit.



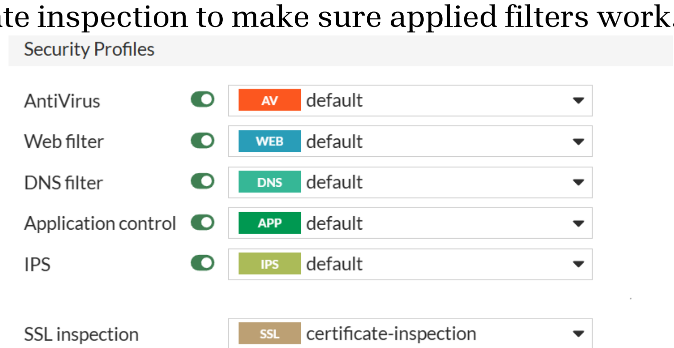

Rename the policy and copy the configurations below, this allows only the specified addresses to access the internet.



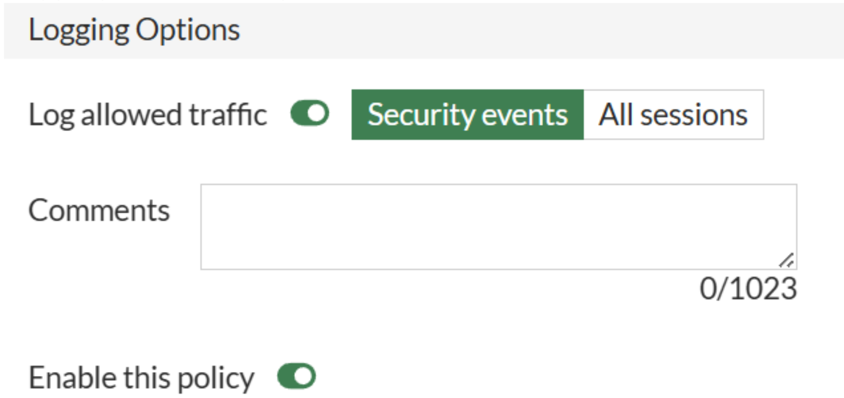

Under Firewall/Network Options ensure NAT is on and the outside interface is being used as outside IP addresses rather than a pool, to preserve the source port and keep the default protocol.




Under Security Profiles, enable everything and select the default profile and enable SSL certificate inspection to make sure applied filters work.

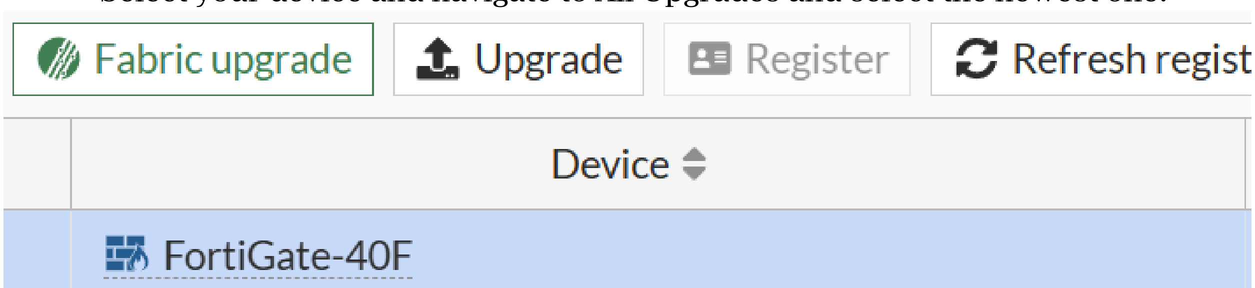


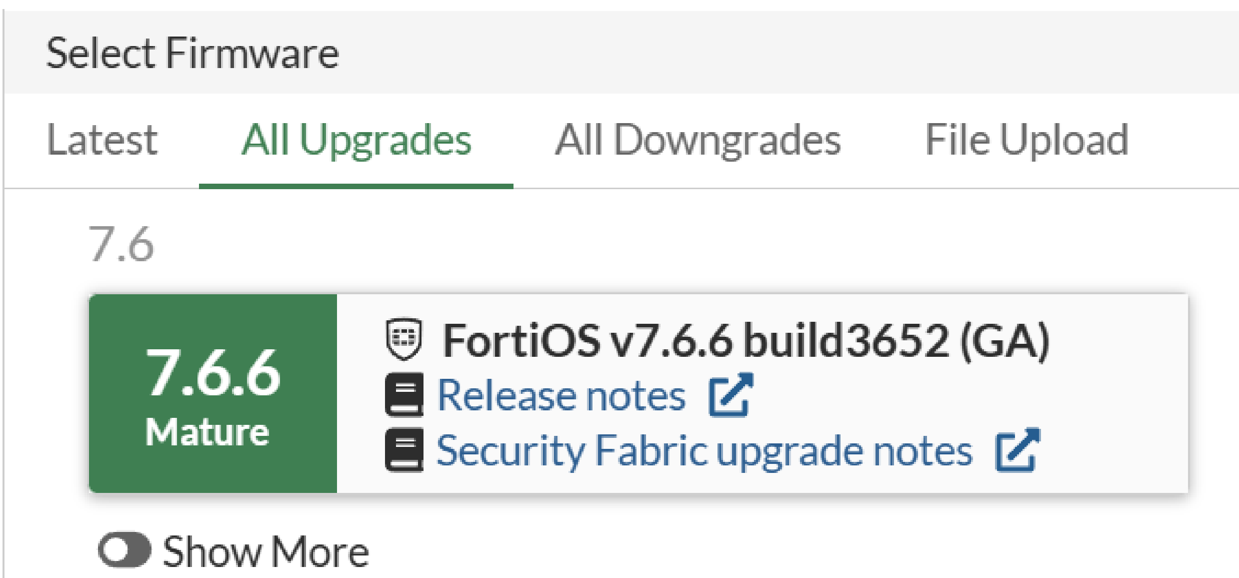
Under Logging Options, log allowed traffic and enable to policy.




Confirm the configurations and now update the firmware to the latest version. Update the firmware by navigating to System > Firmware and Registration in the left navigation panel.

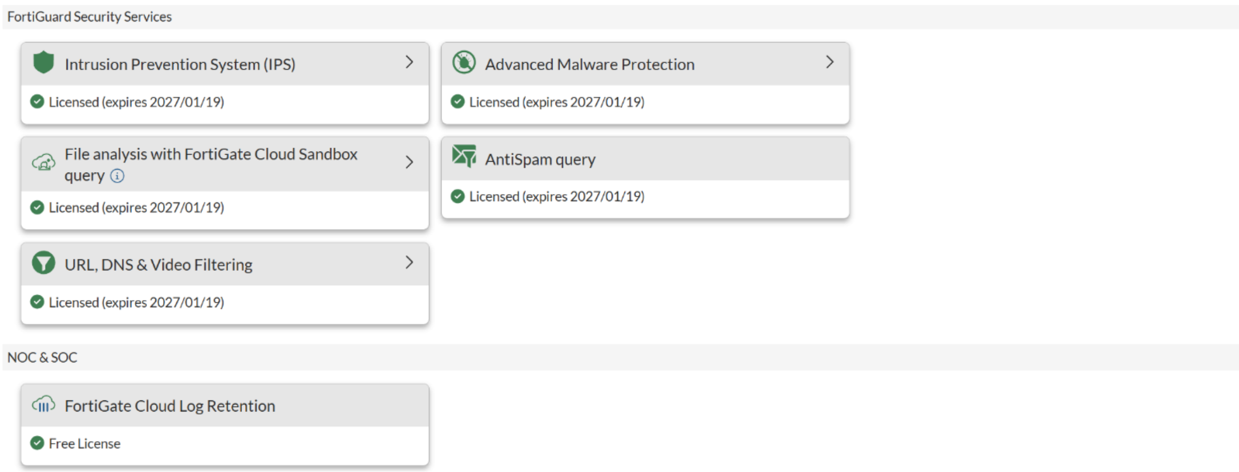
Select your device and navigate to All Upgrades and select the newest one.

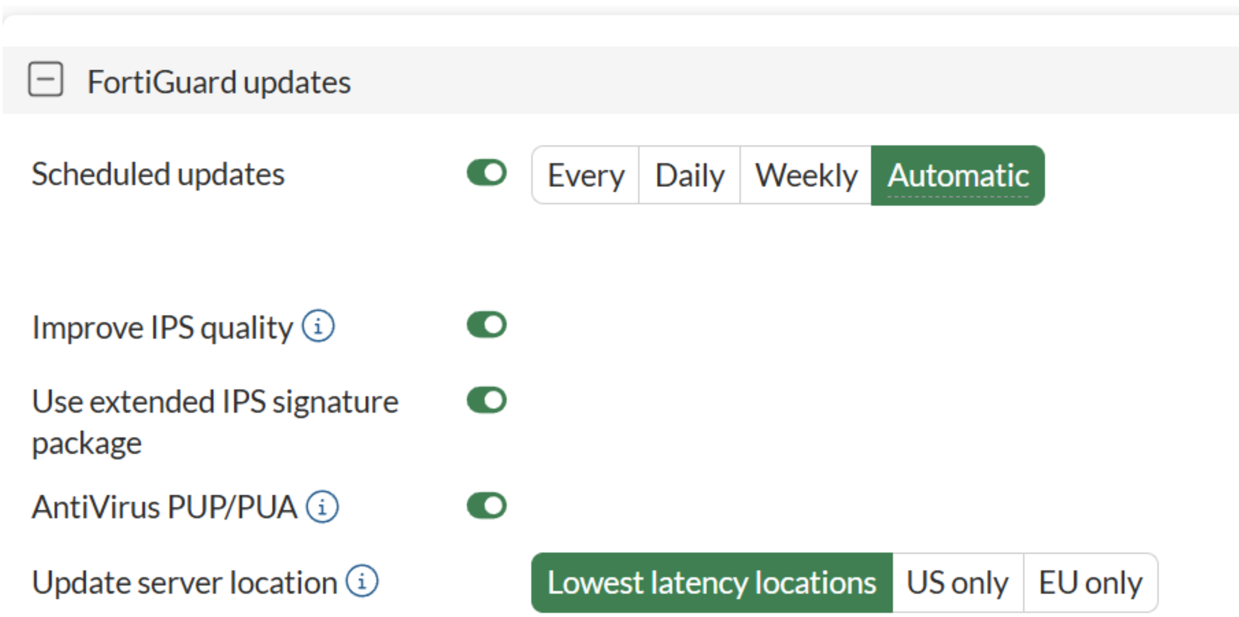



The firewall will have to reboot, once rebooted log back into the firewall.

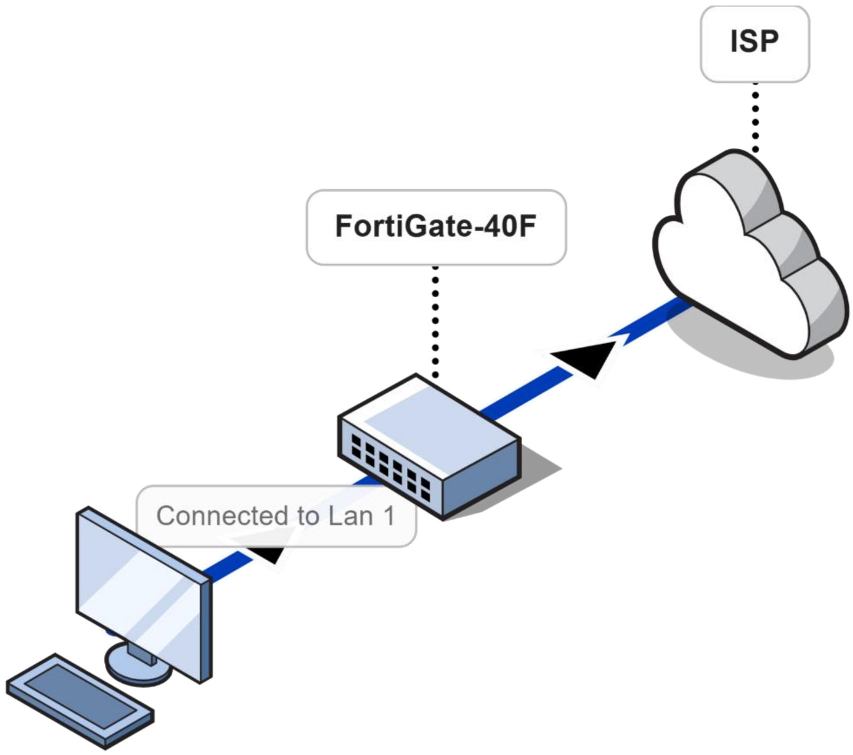
Now in the left Navigation panel under Setting go to FortiGuard to make sure all licenses are up to date and FortiGuard updates are set to be as often as possible.




You have configured the FortiGate-40F with some basic security upgrades for a SOHO environment.

**Network Diagram**




**Problems**

The only problem regarding this lab I had was trying to Factory Reset the firewall, initially we tried to press the button before the firewall had completely booted up so the firewall would not reset. I then unplugged and re-plugged the firewall multiple times until I found out that I had to wait until the firewall had to completely boot. To know if the firewall was booted I consoled into the firewall and waited to press the button until the firewall had booted, even after knowing this it still took a couple tries before being able to factory reset the firewall.

**Conclusion**

This lab helped me gain an understanding of how Fortinet systems operate and their unique advantages that set them apart from their competitors. It also helped to develop an understanding of how security profiles and policies all converge together to create a secure connection to the internet. This lab was a very foundational and basic lab and the skills developed within the lab will greatly transfer to the next lab of creating a WLAN or Wireless Local Area Network that can be accessed from anywhere the firewalls signal can reach instead of cables.